Fiscal Year:	FY 2012	**Task Last Updated:**	FY 06/30/2012
PI Name:	Hasenstein, Karl Ph.D.		
Project Title:	Microgravity and Magnetic Fields - Directional Root Growth and Gene Expression		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	hasenstein@louisiana.edu	**Fax:**	FY 337-482-5834
PI Organization Type:	UNIVERSITY	**Phone:**	337-482-6750
Organization Name:	University of Louisiana - Lafayette		
PI Address 1:	PO Box 42451		
PI Address 2:	300 E. St. Mary Boulevard		
PI Web Page:	http://www.ucs.louisiana.edu/		
City:	Lafayette	**State:**	LA
Zip Code:	70504-2451	**Congressional District:**	3
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2009 Space Life Sciences-Flight NNH09ZTT002N
Start Date:	08/03/2010	**End Date:**	12/31/2013
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	4	**No. of Master' Degrees:**	1
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA KSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	ISS		
Flight Assignment:	NOTE: New end date is 12/31/13 per NSSC information (Ed., 8/6/13)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX10AP91G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This proposal combines morphological measurements on micro-gravity germinated and grown (Brassica) roots that are exposed to high-gradient magnetic fields (HGMF) with a high-resolution spatial analysis of genes involved in gravitropism. Magnetic gradients exert a repulsive force on diamagnetic substances such as the starch in amyloplasts, which results in gravitropic-like curvature. The proposal is based on hardware that was developed for a previous flight experiment on STS-107. As in the originally proposed experiment, microscopic analysis of amyloplast size and distribution will be studied in the context of achieved curvature. In addition, high-resolution genetic profiling will be conducted that focuses on the cytoskeleton and starch development. The result of the proposed work will enable directional growth of seedlings and eventually mature plants under microgravity conditions using HGMF. Three conditions will be used to differentiate microgravity from uniform magnetic fields and HGMF conditions. Only HGMF is expected to influence root curvature. A uniform magnetic field will serve as control for possible secondary magnetic effects. This condition will be compared to an in-flight control without magnetic field to determine growth rate, gene expression, growth, and directionality (autotropism). Root material from these experiments will be used for investigations on cytoskeletal organization and the expression of starch-related genes. A second aspect of the proposed work will investigate the process of gravisensing, whether dependent on amyloplast position, movement, or force exerted onto amyloplasts. Space experimentation is necessary to establish a threshold value for a directional magnetic force. In addition to providing insight in the fundamental mechanism of the gravisensing system of plants, the data from this experiment can be used to develop technology to provide a directional, gravity-like stimulus to plants during early stages of germination. The proposed experiment can be performed with existing hardware, be adapted easily to changing flight schedules, and requires minimal crew time. The anticipated benefits include a better understanding of root development, effects of amyloplast displacement on root development and curvature, and compensatory measures to alleviate microgravity conditions. A newly developed gene extraction technology is ideally suited for gene-profiling studies and has far-reaching implications for studying gene expression, especially for confined space flight experiments. Additional benefits of the project include training of students at the undergraduate and graduate levels.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	In addition to an improved understanding of basic biological questions, such as identification of the site of graviperception, curvature-induced (but not gravity-dependent) gene expression, the experiment also tests the effect of magnetic fields on plant roots. Therefore, biological questions are investigated in conjunction with physical principles. We intent to analyze the effect of static magnetic fields on biological systems. This work will provide a starting point for the current discussion of the detrimental effects of (electro)magnetic fields. In addition, understanding basic biological questions will improve the prospect for enhanced plant productivity, resource optimization, and reduced maintenance.
Task Progress:	The activities in 2011/2012 included two aspects: furthering the understanding of the biology of Brassica seeds and seedlings in the PI's lab and preparations for test at KSC. The activities related to science verification and biocompatibility test were performed at KSC and UL Lafayette. After extensive preparation including optimization of watering volume, fixation times, and storage conditions, a science verification test (SVT) was attempted in Oct 2011. Despite numerous tests in the PI's lab where germination and root elongation occurred even after 12 week storage in Magnetic Field Chambers (MFCs), the germination rate at KSC during the attempted SVT was zero and prompted a massive investigation concerning the root cause of the failed SVT. Since a likely inhibitor of germination includes undesirable levels of organic volatile compounds (VOCs), an analysis of the gas phase was attempted and resulted in elevated levels of ethylene, CO2, and benzyl butyl phthalate. The levels varied between magnetic and non-magnetic chambers and suggested that the mounting of the magnets, the securing of magnets with RVT (silicon compound), recent spray-painting of covers, and/or washing procedures are possible causes for the lack of germination. The extended exposure in a triple sealed container the Generic Containment Unit (GCU) further complicated the precise identification of the problem. Subsequent tests attempted to identify the contributions of all parts of the MFCs to which the seeds were exposed. Testing of two types of RVT (3145 & 3140), gasket and Lexan material as well as tubing and paper turned out negative. Possible differences between the two types of MFCs (legacy and newly made) were explored but were inconclusive. A possibility (still under investigation) includes imprecise seating of the fill plugs that could have formed incomplete seals and thus allowed a higher gas exchange than the new chambers. The situation was complicated further by an unexpected event resulting in the magnetic attachment of the new MFCs. As a result, the PI's MFCs were shipped to KSC and used in subsequent tests. The joined MFCs still await attempts to separate them. After introducing activated carbon cloth (ACC), the germination rate improved dramatically. This observation resulted in a modification to the MFCs that includes ACC for future tests. One test with storage of 23 d prior to imbibition resulted in germination but shorter than expected root length. Further tweaking of the amount of ACC and activation procedure is likely to result in germination rates that were obtained in fresh controls. With the progress in germination, optimization of fixative and storage conditions, a second SVT is scheduled to begin July 12, 2012. Research activities in the PI lab dealt with related problems: the storability of specimen in RNALater, the longevity of formaldehyde-based fixative in the presence of corroding NdFeB magnets, immune-staining procedures, and the detectability of gene expression changes associated with germination, gravitropic curvature, and growth. In addition, related research studied the effect of mechanostimulation on the formation of apoptotic cells in the Brassica columella. RNALater studies showed that the custom preparation of the fixative was superior to the commercially available product in long-term experiments. However, the process of fixation seems to degrade RNA before the preservative effects are fully realized. Current research is trying to determine if the loss of apparent quality is the result of mRNA or ribosomal RNA degradation. Studies with a RNA analyzer (Labchip 90) showed that both fractions are affected but that mRNA is less degraded than the total RNA fraction. The delivered and soon to be activated Qiagen system (QIAxcel RNA and DNA analyzer) will confirm the sensitivity of fresh material to the (slow) exposure to RNALater. So far, the results of these studies indicate significant variability to the exposure to RNALater but not the duration of preservation of the biological material (whole brassica seedlings). One problem for subsequent processing of preserved RNA is the removal of the high salt concentration for down-stream analysis (Q-PCR, agarose electrophoresis etc.). The storage conditions of formaldehyde fixed material in corroded NdFeB magnet material was tested by storing formaldehyde-fixed material in

	the presence of magnet fragments. The discoloration of the solution was significant. However, despite a tougher (leathery) of root tissue, subsequent immune-staining was relatively unaffected. These data indicate that corrosion mainly affects the surface but not the internal cellular structure of root cells. The testing of Q-PCR-based gene-expression analyses showed that RNALater-fixed material is suitable for gene-expression studies. We are still 'calibrating' the expression data against the effect of RNALater fixation. Studies of apoptosis as a consequence of mechanostimulation are relevant because of the reduction of mechanostimulation in space environment (only internal, cellular events will affect mechanosensors). Clinorotation (substitute but reproducible mechanostimulation) showed enhanced sensitivity of roots to stimuli (enhanced curvature with higher angular velocity). The next step of this analysis will include gene expression studies after clinorotation. More refined investigations of the organization of the root cap cytoskeleton (based on a new plug-in for ImageJ, OrientationJ) is suitable for cytoskeletal orientation analysis of individual cells and can reveal the correlation between f-actin and tubulin organization.
Bibliography Type:	Description: (Last Updated: 01/20/2026)
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Abstracts for Journals and Proceedings	Park MR, Hasenstein KH. "Expression Profiling of Genes in Brassica Seedlings." 27th Annual Meeting of the American Society for Gravitational and Space Biology, San Jose, CA, November 3-6, 2011. Program and abstracts. 27th Annual Meeting of the American Society for Gravitational and Space Biology, San Jose, CA, November 3-6, 2011. p. 104-105. , Nov-2011
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